

Program : Diploma in Information Technology	
Course Code : 4262	Course Title: System Administration
Semester : 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To provide the basic concepts of system administration logics which will help to create and manage computer system, manage user accounts, managing disk and file system.

Course Pre-requisites:

Topic	Course Code	Course Title	Semester
Basic knowledge of partitioning		Computer Hardware and Networking Lab	3
Basic Knowledge of programming		Fundamentals of System Administration	3

Course Outcomes:

On completion of the course, students will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate the basic concepts of system administration, administration tools and different partition types	15	Understanding
CO2	Outline the basic concepts of managing software, RPM and YUM	14	Understanding
CO3	Summarize the concepts of managing user account, create user account and create group account	15	Understanding
CO4	Summarize the concept of managing disk and file system	14	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate the basic concepts of system administration, administration tools and different partition types		
M1.01	Summarize system administration	1	Understanding
M1.02	Outline ‘su’ and ‘sudo’ command	1	Understanding
M1.03	Summarize fedora from live media	1	Understanding
M1.04	Summarize Red Hat enterprise Linux	1	Understanding
M1.05	Explain graphical administration tool	1	Understanding
M1.06	Classify the graphical tools in fedora and Red Hat enterprise	1	Understanding
M1.07	Explain services, authentication, date and time, firewall, language, printing, user and groups	2	Understanding
M1.08	Explain dual booting	1	Understanding
M1.09	Explain how to partitioning hard drives	1	Understanding
M1.10	List the different partition types	1	Remembering
M1.11	Explain Linux partition	1	Understanding
M1.12	Explain LVM partition and RAID partition	2	Understanding
M1.13	Illustrate partition to directories	1	Understanding
Contents: System administration-su command-sudo command-installing fedora from live media- installing RedHat enterprise Linux-graphical administration tool-graphical tool in fedora and RedHat enterprise- (services, authentication, date and time, firewall, language, printing, user and groups)dual booting-partitioning hard drives-different partition types			

(Linux partition, LVM partition, RAID partition, swap partition)-partition to directories			
CO2	Outline the basic concepts of managing software, RPM and YUM		
M2.01	Outline RPM	1	Understanding
M2.02	Explain Linux RPM	1	Understanding
M2.03	Explain DBE software packaging	1	Understanding
M2.04	Explain RPM packaging	1	Understanding
M2.05	Outline Where do RPM come from	1	Understanding
M2.06	Explain how to install and uninstall RPM package	1	Understanding
M2.07	Illustrate Yum configuration and explain basic Yum commands	1	Understanding
M2.08	Explain different installation modes of Yum	1	Understanding
M2.09	Demonstrate the methods to installing software in Linux	1	Understanding
M2.10	Explain how YUM works	1	Understanding
M2.11	Illustrate YUM with third-party software repositories.	1	Understanding
M2.12	Explain managing software with YUM command	1	Understanding
M2.13	Explain removing package using YUM	1	Understanding
M2.14	Explain installing, Querying and verifying software with rpm command	1	Understanding
	Series Test – I	1	
Contents: What is RPM-Linux RPM- DBE software packaging – RPM packaging -Where do RPM come from- Installing RPM, Install and uninstall RPM package--What is YUM, Illustrate Yum configuration and basic Yum commands, installation modes of Yum, Method installing software in Linux, understanding how YUM work- Understanding YUM with third-party software repositories, managing software with YUM command -Removing package using YUM - Installing, Querying and verifying software with rpm command.			
CO3	Summarize the concepts of managing user account, create user account and create group account		
M3.01	Outline create user account in Linux	1	Understanding
M3.02	Explain User account field	1	Understanding
M3.03	Explain account type, full name, user name, password, verify	2	Understanding
M3.04	Summarize Managing user in the Enterprise	1	Understanding
M3.05	Illustrate usermod commands with example	1	Understanding

M3.06	Explain using group account	2	Understanding
M3.07	Summarize creating group account	1	Understanding
M3.08	Outline access control list (ACL)	1	Understanding
M3.09	Explain Setting permissions with ACL	2	Understanding
M3.10	Compare UID and GID	1	Understanding
M3.11	Summarize set UID and GID bit commands	1	Understanding
M3.12	Explain centralizing user accounts using User window and authentication configuration window	1	Understanding
Contents: Managing user account-create user account in Linux-setting user default-usermod commands- example of usermod command-deleting user with userdel-using group account-creating group account- Managing user in the Enterprise-access control list (ACL)- Difference between UID and GID-How to set UID and GID bit commands-Centralizing User accounts- User window and authentication configuration window.			
CO4	Summarize the concept of managing disk and file system.		
M4.01	Summarize disk storage	1	Understanding
M4.02	Explain how to partitioning hard disk	1	Understanding
M4.03	Explain partition table	1	Understanding
M4.04	Summarize viewing disk partition	1	Understanding
M4.05	Explain how to partition a disk with single	1	Understanding
M4.06	Explain how to partition a disk with multiple	1	Understanding
M4.07	Outline LVM	1	Understanding
M4.08	Explain logical volume management partitioning	1	Understanding
M4.09	Outline file system	1	Understanding
M4.10	Summarize create new file system	2	Understanding
M4.11	Explain Mounting file system	2	Understanding
M4.12	Explain how to create a file system using mkfs command	1	Understanding
	Series Test – II	1	
Contents: Disk storage, partitioning hard disk, understanding partition table, viewing disk partition, creating a single partition disk, general procedure in single partitioning, partition a disk with multiple, what is LVM, file system, mounting file system, how to create new file system, create a file system using mkfs command.			

Text / Reference:

T/R	Book Title / Author
T1	Christopher Negus “Linux Bible” the comprehensive tutorial resources, Ninth Edition
T2	Richard Petersen “Linux: The Complete Reference”, Sixth Edition
R1	Michael Jang, Alessandro Orsaria “RHCSA/RHCE Red Hat Certification Study Guide”, Seventh Edition

Online Resources:

Sl.No	Website Link
1	https://www.tuxcademy.org/download/en/adm1/adm1-en-manual.pdf
2	https://www.tecmint.com/20-linux-yum-yellowdog-updater-modified-commands-for-package-mangement/
3	https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/6/html/deployment_guide/s2-users-cl-tools